GNN-Based Link Prediction on the LUCIA Knowledge Graph

Results Summary - April 2026

Setup

Graph: 22,553 nodes, 177,712 edges, 13 node types, 20 relation types. Three models compared: Dot Product, TransE, R-GCN with DistMult decoder. 128-dim embeddings, Adam (LR 0.001), 200 epochs, batch 4,096, 10 negatives per positive. 80/10/10 split. Filtered ranking, 500 test triples per run.

Evaluation Metrics

For each test triple (h, r, t), the model scores every possible tail entity in the graph and ranks them by score. The rank of the correct tail is recorded. Two standard metrics are then computed:

Mean Reciprocal Rank (MRR): Average of 1/rank across all test triples. If the correct entity is always ranked first, MRR = 1.0. If it is on average ranked tenth, MRR = 0.1. Higher is better.
Hits@k: Proportion of test triples where the correct entity appears in the top k predictions. Hits@10 = 0.85 means the correct entity is in the top 10 in 85% of cases. Reported for k = 1, 3, 10.

All evaluations use the filtered ranking protocol: known true triples (other than the test triple itself) are removed from the candidate set so that the model is not penalised for ranking valid but unobserved triples highly.

1. Ablation Study: Does Environmental Data Help Gene-Disease Prediction?

The same R-GCN architecture was trained on three different graph subsets to isolate the contribution of environmental data. **Combined** uses both biological and environmental edges. **Bio-only** removes environmental edges and the subtype_of bridges. **Env-only** keeps only environmental edges plus the subtype_of bridges, so it serves as a sanity check that environmental data alone cannot predict gene-disease relationships (which by definition it should not, since environmental data has no direct gene connections).

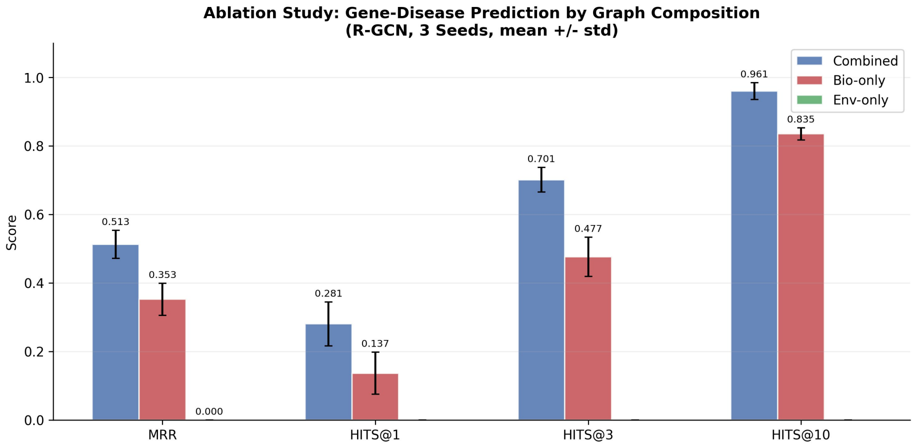

Figure 1: R-GCN trained on three different graph subsets (3 seeds each).

Configuration	Train Triples	MRR	Hits@1	Hits@3	Hits@10
Combined (Bio + Env)	142,170	0.513	0.281	0.701	0.961
Bio-only	~64,000	0.353	0.137	0.477	0.835
Env-only	~78,000	0.000	0.000	0.000	0.000

Table 1: Ablation study on Gene-Disease prediction (mean across 3 seeds, seeds 42/123/456). Adding environmental data improves MRR by 45% (0.353 to 0.513) and Hits@10 by 15 percentage points (0.835 to 0.961). The Env-only row (greyed out) confirms that environmental data alone has no predictive power for gene-disease relationships.

2. Gene-Disease Link Prediction Across Models (5 seeds)

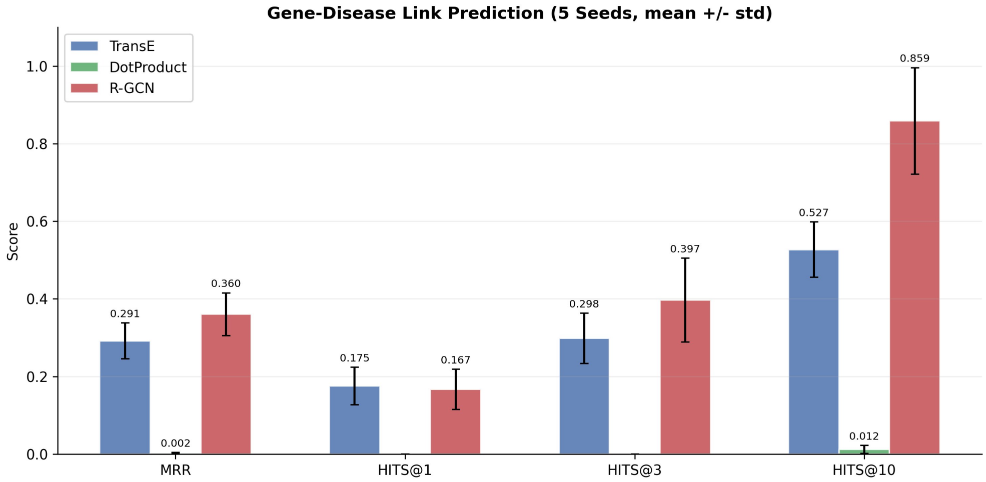

Figure 2: Gene-Disease link prediction across 5 random seeds.

Model	MRR	Hits@1	Hits@3	Hits@10
R-GCN	0.360 ± 0.055	0.167 ± 0.054	0.397 ± 0.107	0.859 ± 0.137
TransE	0.291 ± 0.046	0.175 ± 0.048	0.298 ± 0.065	0.527 ± 0.071
Dot Product	0.002 ± 0.002	0.000 ± 0.000	0.000 ± 0.000	0.012 ± 0.010

Table 2: Multi-seed Gene-Disease link prediction. R-GCN leads on MRR and Hits@10. Dot Product collapses to near zero, confirming that entity embeddings without relation modelling cannot capture gene-disease patterns.

3. Overall Link Prediction (across all 20 relation types)

Model	MRR	Hits@1	Hits@3	Hits@10
TransE	0.568 ± 0.020	0.499 ± 0.024	0.594 ± 0.026	0.669 ± 0.020
R-GCN	0.465 ± 0.012	0.378 ± 0.014	0.510 ± 0.014	0.568 ± 0.012
Dot Product	0.115 ± 0.014	0.071 ± 0.011	0.122 ± 0.018	0.180 ± 0.024

Table 3: Overall link prediction across all 20 relation types.

Note: TransE leads on overall MRR primarily because of its strong performance on the Gene-Pathway relation (147 test triples, ca. 30% of all test cases). R-GCN dominates on the smaller, more structured relations such as Gene-Disease (see Section 4).

4. R-GCN Performance per Relation Type (test sample n ≥ 10)

Per-relation MRR and Hits@10 for R-GCN. Only relations with at least 10 test triples are shown to ensure statistically meaningful values. Relations with smaller test samples are omitted here; the AUROC analysis in Section 5 provides a more robust assessment based on the full edge set.

Relation	n test	MRR	Hits@10
Disease → cancer_stats_in → Country	26	1.000	1.000
GeographicRegion → measured_in → CalendarYear	27	0.975	1.000
GeoPoliticalRegion → measured_in → CalendarYear	98	0.919	1.000
Gene → associated_with → Disease	**42**	**0.602**	**0.976**
Disease → cancer_stats_in → GeographicRegion	74	0.595	1.000
GeoPoliticalRegion → part_of → Country	11	0.348	0.636
Chemical → exposure_in → GeoPoliticalRegion	36	0.094	0.278
Gene → in_pathway → Pathway	147	0.013	0.020

Table 4: R-GCN per-relation performance (single run, seed 42), restricted to relations with n ≥ 10 test triples. Gene-Disease relation highlighted.

5. Score Calibration: How Well Does the Model Distinguish True from False?

The DistMult scoring function produces unbounded real-valued scores. To assess how well the trained R-GCN distinguishes true from false triples, we generate 1,000 random false triples per relation type (type-aware sampling, excluding all known true triples) and score them with the trained model. We then compare this distribution against the distribution of all known true triples in the graph.

TRUE median: Median score that the model assigns to known true triples of this relation (triples that exist in the graph). **FALSE median:** Median score that the model assigns to the 1,000 random false triples of this relation. The gap between TRUE median and FALSE median indicates how strongly the model separates the two classes. **AUROC:** Probability that a random known-true triple receives a higher score than a random false triple. AUROC = 1.0 means perfect separation; AUROC = 0.5 means random guessing.

Relation	TRUE median	FALSE median	AUROC
Disease → cancer_stats_in → GeographicRegion	33.530	-5.348	1.000
Disease → cancer_stats_in → Country	18.863	-3.594	1.000
Chemical → exposure_in → GeographicRegion	7.394	-2.069	0.994
GeographicRegion → measured_in → CalendarYear	5.058	-3.607	0.993
Biomarker → marker_for → Disease	0.623	0.206	0.992
Disease → has_rearrangement → ChromoRearr	3.724	1.026	0.990
Pathway → linked_to → Disease	1.309	0.209	0.987
Disease → has_fusion → GeneFusion	2.847	-0.063	0.986
Disease → subtype_of → Disease	5.468	0.478	0.985
Variant → variant_of → Disease	0.532	-0.295	0.983
GeographicRegion → cancer_data_in → CalendarYear	7.960	-2.580	0.976
GeoPoliticalRegion → measured_in → CalendarYear	2.544	-7.691	0.972
Variant → located_in_gene → Gene	0.118	-0.533	0.957
Gene → associated_with → Disease	**2.185**	**0.201**	**0.933**
Chemical → exposure_in → GeoPoliticalRegion	2.602	0.002	0.899
Gene → in_pathway → Pathway	2.042	0.095	0.868
Country → cancer_data_in → CalendarYear	13.277	5.489	0.865
GeneProduct → part_of_pathway → Pathway	0.321	0.198	0.820
GeoPoliticalRegion → part_of → Country	3.093	1.476	0.803
GeographicRegion → part_of → Country	6.568	6.102	0.601

Table 5: AUROC of R-GCN scores against random false triples per relation. Sorted by AUROC. The Gene-Disease relation (highlighted) achieves AUROC 0.933.

6. Score Distributions: True vs False vs Top Predictions

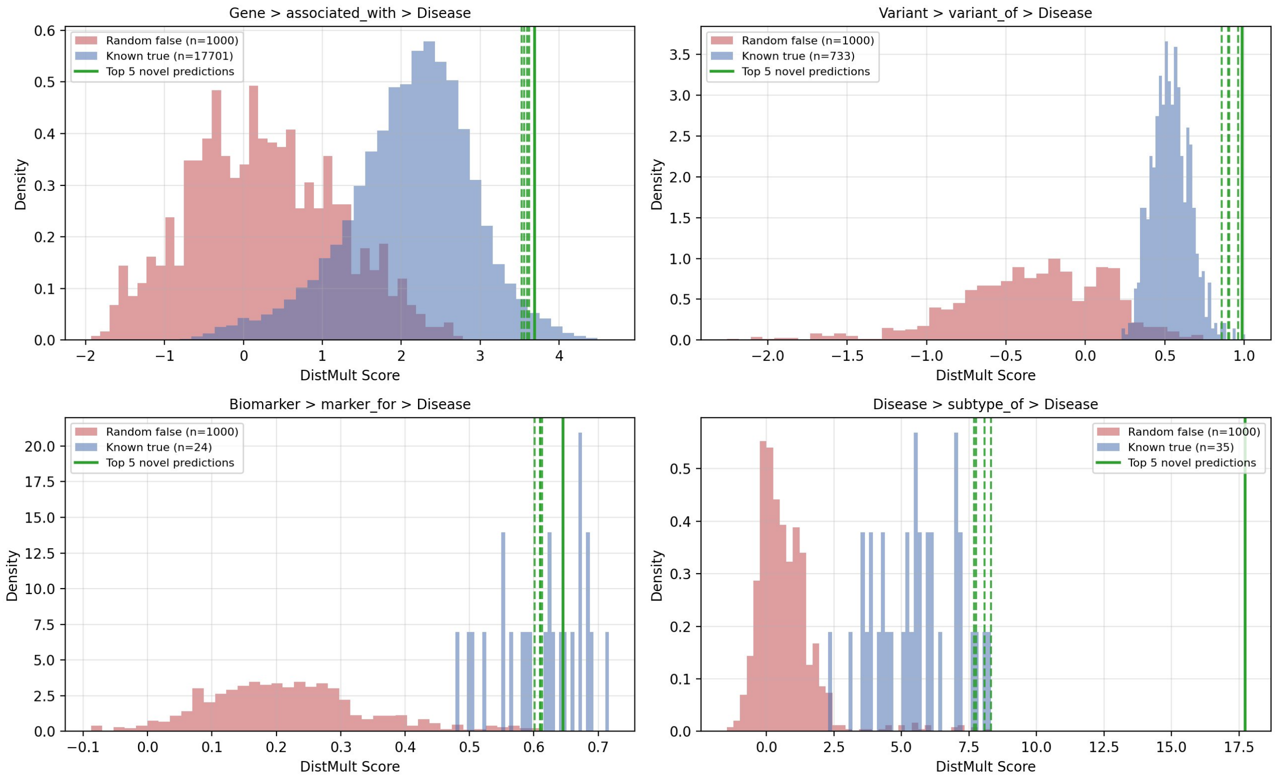

Figure 3: Score distributions for four key relations. Red: 1,000 random false triples. Blue: all known true triples. Green vertical lines: top 5 novel predictions.

Reading the plot: Where the red and blue distributions are clearly separated (Variant-Disease, Biomarker-Disease, Disease-subtype), the model can reliably distinguish true from false. Where they overlap more (Gene-Disease has some overlap in the 1.0-2.5 score range), the model is less certain. The green lines mark the top 5 novel predictions: in all four panels they sit at or above the high end of the true triple distribution and well above the false distribution.

7. Novel Predictions with Confidence Validation

Confidence labels combine two checks against the score distributions from Section 5: **very strong** = score above 99% of random false triples AND above 75% of known true triples. **strong** = above 95% false AND above 50% true. **moderate** = above 80% false AND above 25% true. **weak** = otherwise.
The "**vs TRUE**" column shows the percentage of known true triples (of the same relation) that score lower than this prediction. A value of 98% means the prediction scores higher than 98% of the known true triples in the graph, putting it in the top 2% of the model's beliefs about this relation. The "vs FALSE" check is implicit in the confidence label (it must be 100% or above 99% for "very strong" / "strong").

7.1 Top Novel Gene-Disease Predictions (n_true = 17,701)

Rank	Gene	Predicted Disease	Score	vs TRUE	Confidence
1	CASP8	Squamous cell carcinoma of lung	3.689	98%	very strong
2	TLR9	Squamous cell carcinoma of lung	3.622	98%	very strong
3	KDR	Squamous cell carcinoma of lung	3.593	98%	very strong
4	PIK3R2	Squamous cell carcinoma of lung	3.521	97%	very strong
5	EGFR	Squamous cell carcinoma of lung	3.476	96%	very strong

Table 6: Top 5 novel Gene-Disease predictions. All score in the top 4% of the 17,701 known true triple distribution.

7.2 Top Novel Variant-Disease Predictions (n_true = 733)

Rank	Variant	Located in	Disease	Score	vs TRUE	Confidence
1	rs1436873982	XRCC1	SCLC	0.986	>99%	very strong
2	rs1057519747	TP53	NSCLC	0.962	>99%	very strong
3	rs121913343	TP53	NSCLC	0.904	>99%	very strong
4	rs397517114	ALK	NSCLC	0.853	99%	very strong
5	rs755024903	NLRP2	NSCLC	0.821	98%	very strong

Table 7: Top 5 novel Variant-Disease predictions. SCLC = Small Cell Lung Carcinoma; NSCLC = Non-Small Cell Lung Carcinoma.

7.3 Top Novel Biomarker-Disease Predictions (n_true = 24)

Rank	Biomarker	Predicted Disease	Score	vs TRUE	Confidence
1	Selenium	Squamous cell carcinoma of lung	0.645	63%	strong
2	Glycitein	Squamous cell carcinoma of lung	0.613	42%	moderate
3	Selenium	Adenocarcinoma of lung	0.610	42%	moderate
4	Glycitein	Adenocarcinoma of lung	0.610	42%	moderate
5	Selenium	Lung Adenocarcinoma In Situ	0.601	42%	moderate

Table 8: Top 5 novel Biomarker-Disease predictions. With only 24 known true triples, the distribution is small. Top predictions cluster around the median of known true triples.

7.4 Top Novel Disease Subtype Predictions (n_true = 35)

Rank	Parent Disease	Predicted Subtype	Score	vs TRUE	Confidence
1	Malignant neoplasm of lung	Lung Carcinoma Metastatic	8.309	100%	very strong
2	Malignant neoplasm of lung	Non-Small Cell Lung Carcinoma	8.067	97%	very strong
3	Malignant neoplasm of lung	Small cell carcinoma of lung	7.954	94%	very strong

Table 9: Top 3 novel Disease subtype predictions. With only 35 known subtype triples, the small distribution makes "100% vs TRUE" easier to achieve, but the predictions are still consistent with established lung cancer taxonomy.

8. Chemical-Disease Embedding Geometry

A permutation test on Chemical-Disease cosine similarities (observed mean 0.558 vs random 0.762 ± 0.037, p = 1.0) shows that the trained R-GCN does **not** encode Chemical-Disease relationships through embedding proximity. The information flow from chemicals to diseases is structural, propagating through Disease hubs via cancer_stats_in and subtype_of edges, rather than geometric. This is consistent with the ablation result in Section 1: environmental data helps predictions through graph structure, not through direct embedding-space alignment.

9. Country-Level Pollutant Exposure vs Lung Cancer Mortality

Pearson correlation between mean country-level pollutant exposure and lung cancer mortality rate. Standard interpretation: |r| < 0.1 negligible, 0.1-0.3 weak, 0.3-0.5 moderate, 0.5-0.7 strong, > 0.7 very strong.

Pollutant	Pearson r	n countries	Strength
Pb (PM10 Lead fraction)	**+0.446**	**24**	**moderate**
BaP (Benzo[a]pyrene)	+0.172	12	weak
As (PM10 Arsenic fraction)	+0.108	24	weak
C6H6 (Benzene)	-0.002	24	negligible
Cd (PM10 Cadmium fraction)	-0.027	24	negligible
Ni (PM10 Nickel fraction)	-0.100	24	negligible
PM10	-0.111	32	weak
PM2.5	-0.144	32	weak
NO2	-0.145	32	weak
O3 (Ozone)	-0.209	24	weak

Table 10: Country-level Pearson correlations. Only Pb (lead) shows a moderate positive correlation. The other pollutants show weak or negligible correlations at this aggregation level, likely due to confounding with smoking prevalence and healthcare access at the national level.

10. Risk Quadrants: PM2.5 and Pb Exposure vs Lung Cancer Mortality

While Section 9 shows Pb has the strongest country-level correlation, PM2.5 is presented here as well because it is the most studied air pollutant in lung cancer epidemiology and classified as IARC Group 1 carcinogen.

10.1 PM2.5 Exposure vs Lung Cancer Mortality

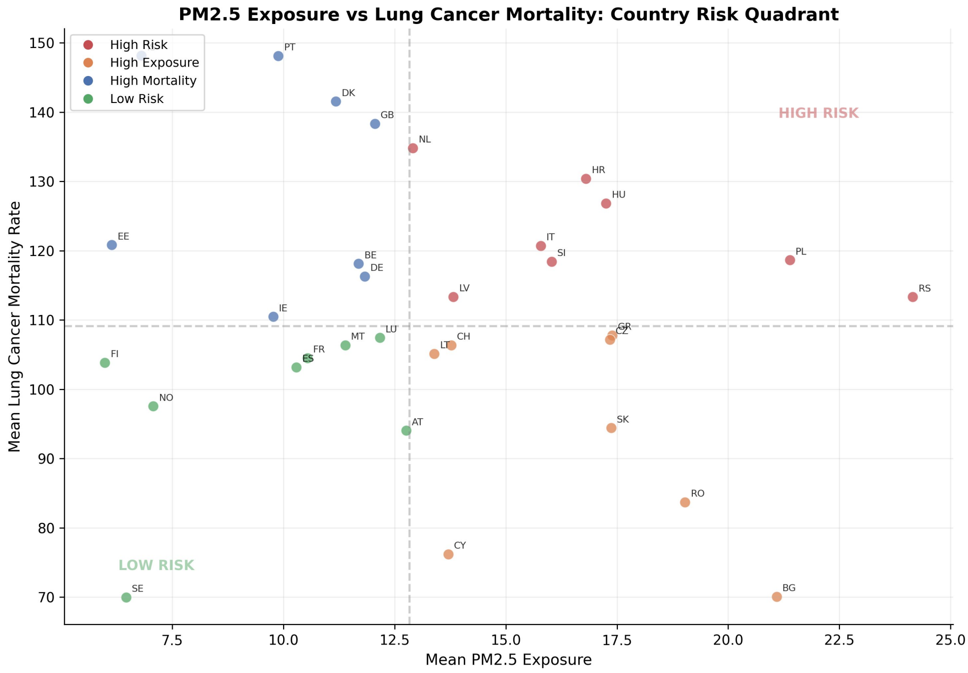

Figure 4: Country-level PM2.5 exposure vs lung cancer mortality. Pearson r = -0.144 (weak). The high-risk quadrant (top right) contains 8 countries.

Country	PM2.5 (ug/m3)	Mortality (per 100k)
Serbia (RS)	24.1	113.3
Poland (PL)	21.4	118.6
Hungary (HU)	17.3	126.8
Croatia (HR)	16.8	130.4
Slovenia (SI)	16.0	118.4
Italy (IT)	15.8	120.7
Latvia (LV)	13.8	113.3
Netherlands (NL)	12.9	134.8

Table 11: Eight countries in the high-PM2.5 / high-mortality quadrant.